Catalogue of Catalogues: A Web Platform for Earthquake Catalogue Management, Merging, and Statistical Analysis

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Abstract

We present the *Catalogue of Catalogues* (CofC), an open-source web platform for ingesting, quality-assessing, merging, and statistically analysing earthquake catalogues from multiple agencies and formats, designed in accordance with the FAIR data principles (Findable, Accessible, Interoperable, Reusable). CofC accepts data as CSV/TXT, GeoJSON, QuakeML 1.2 Basic Event Description (BED), and live FDSN web-service streams. A five-strategy merging engine detects duplicate events using configurable time ($\leq 60\,\mathrm{s}$), distance ($\leq 50\,\mathrm{km}$), and magnitude (≤ 0.5) thresholds and resolves conflicts via priority, quality-score, newest-data, most-complete, or averaged-value rules. An objective quality-scoring system (0–100 index, A^+ to F letter grade) evaluates each event across five dimensions: field completeness, physical-consistency accuracy, instrumental reliability, inter-field consistency, and processing traceability. Standard seismological analyses—Gutenberg-Richter b-value estimation via maximum likelihood, magnitude of completeness (M_c) via the Maximum Curvature (MAXC) method, Gardner-Knopoff and Reasenberg window declustering, and cumulative seismic moment—are available in the browser. Using synthetic New Zealand seismicity data representative of a two-catalogue merge, we demonstrate the core workflow: 50% of events in a secondary agency catalogue were identified as duplicates of the primary catalogue, and quality scoring revealed systematic azimuthal-gap deficiencies in offshore events; after merging and quality filtering, subsequent Gutenberg-Richter analysis yields a stable $\hat{b} = 1.02 \pm 0.03$ with full provenance retained throughout. Six categories of statistical bias affecting merged-catalogue analysis are identified and actionable mitigation guidance is provided. Source code and documentation are freely available under the MIT licence.

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1 Introduction

1.1 The role of earthquake catalogues in seismology

Earthquake catalogues are the fundamental observational basis for quantitative seismology. Every tool in the modern seismological toolkit—from probabilistic seismic hazard analysis (PSHA; Cornell 1968) to operational earthquake forecasting (Schorlemmer et al., 2007) to tectonic geodesy—rests on the assumption that the underlying catalogue faithfully represents the space-time-magnitude distribution of seismicity. The Gutenberg-Richter (GR) frequency-magnitude relation (Gutenberg and Richter, 1944) and its b-value, which encodes the relative proportion of small to large earthquakes (Kagan, 1999; Schorlemmer et al., 2005), are estimated directly from catalogue data; any systematic error in the catalogue propagates directly into hazard and risk assessments. Similarly, the Epidemic-Type Aftershock Sequence (ETAS) model (Ogata, 1988), the most widely used stochastic framework for short-term earthquake forecasting, requires a complete, Poissonian background catalogue that is free from both incompleteness and aftershock contamination.

Regional monitoring agencies—GeoNet (New Zealand; Petersen et al. 2011), the Northern California Earthquake Data Center (Northern California Earthquake Data Center, 2014), the International Seismological Centre (ISC; Storchak et al. 2013), the Engdahl-van der Hilst-Buland relocated global catalogue (Engdahl et al., 1998)—each maintain catalogues with overlapping geographic coverage but different station networks, velocity models, magnitude scales, analysis procedures, and update frequencies. Researchers routinely need to *merge* or *compare* these sources: to extend a short modern catalogue with historical data, to supplement a global catalogue with a higher-resolution regional one, or to construct a unified input for PSHA over a region monitored by multiple agencies. This multi-source workflow introduces three classes of problem that, if ignored, produce statistically invalid results.

1.2 Problem 1: Format heterogeneity

No universally adopted exchange format exists for seismological data. The QuakeML 1.2 Basic Event Description (BED) schema (Schorlemmer et al., 2011) provides a comprehensive XML vocabulary—covering origins, magnitudes, focal mechanisms, picks, arrivals, and amplitude measurements—and is the standard output of the FDSN Event Web Service (International Federation of Digital Seismograph Networks, 2013). However, many agency archives remain in agency-specific CSV layouts, plain text bulletins, or legacy binary formats. The GeoJSON format (Butler et al., 2016) has gained traction for web-facing services but provides no controlled vocabulary for seismological fields; column and property names vary freely between providers. When a researcher assembles a multi-source dataset, the practical first step is writing bespoke format-conversion code—a task that is repetitive, error-prone, and rarely shared or version-controlled.

1.3 Problem 2: Duplicate events and location scatter

When two catalogues cover the same region, the same earthquake typically appears in both, reported with slightly different origin times, hypocentres, and magnitudes. These differences arise from genuine physical causes: different station sets lead to different travel-time residuals; regional versus global velocity models bias depth systematically; automatic picks differ from manually reviewed ones. Bondár and McLaughlin (2009) showed that even for well-recorded teleseismic events, agency-to-agency location scatter can reach 10–20 km in horizontal position and 30 km in depth. Naïvely concatenating catalogues without duplicate removal inflates $N(M)$ at all magnitudes above the lower agency's M_c, lowers the apparent a-value, and distorts rate estimates. Storchak et al. (2013) describe the careful duplicate-association procedure developed for the ISC-GEM global catalogue, which required multi-year expert review; automated, configurable duplicate detection for arbitrary catalogue pairs has not previously been available in a production web interface.

1.4 Problem 3: Statistical biases in merged catalogues

Even after format conversion and duplicate removal, several statistical biases require active mitigation before standard analyses are valid.

Magnitude of completeness (M_c). A catalogue is complete above M_c—the smallest magnitude at which essentially all events are detected by the network. Below M_c, the GR relation breaks down (Rydelek and Sacks, 1989). Fitting b without first determining and enforcing M_c leads to systematic underestimation (Woessner and Wiemer, 2005). Multiple methods for estimating M_c have been proposed (Wiemer and Wyss, 2000; Woessner and Wiemer, 2005; Mignan and Woessner, 2012), each with different assumptions and performance characteristics. When merging two catalogues with different M_c values (e.g., $M_c = 1.5$ for a dense regional network and $M_c = 3.0$ for a global catalogue), the combined FMD will be incomplete between the two thresholds unless the analyst enforces the higher cut-off.

Magnitude scale heterogeneity. Different agencies routinely use different magnitude scales: local magnitude (M_L; Gutenberg and Richter 1944), body-wave magnitude (m_b), surface-wave magnitude (M_s), and moment magnitude (M_w; Hanks and Kanamori 1979; Kanamori 1977). Each scale saturates at a different value—M_L above ~ 6.5, m_b above ~ 6.0 (Bormann and Saul, 2009)—and the inter-scale regressions carry substantial scatter (Das et al., 2011). Mixing scales inflates the apparent variance of the FMD, artificially shifts M_c, and can change the estimated b by up to ± 0.2 (Marzocchi and Sandri, 2003). Tinti and Mulargia (1987) showed that even with a single scale, binning width and sample size both affect the precision of $\hat{b}$ in ways that are routinely underappreciated.

Aftershock clustering. Aftershock sequences violate the Poissonian independence assumption underpinning the GR relation and ETAS background estimation. Failing to decluster inflates event counts at magnitudes just above M_c (where aftershocks are most numerous) and biases $\hat{b}$ downward (Gardner and Knopoff, 1974). Multiple declustering algorithms exist—the window method of Gardner and Knopoff (1974) with Uhrhammer (1986) parameters, the link-based method of Reasenberg (1985), and the ETAS-based stochastic method of Zhuang et al. (2002)—with different sensitivities to swarm-type seismicity and to the aftershocks of aftershocks (van Stiphout et al., 2012). Helmstetter et al. (2005) demonstrated that ignoring small-magnitude aftershocks—often the majority of any sequence—underestimates cascading stress transfer and therefore the effective rate of background seismicity. The choice of declustering method and parameters is a significant source of uncertainty that is rarely propagated through to final hazard estimates.

Location quality filtering. Events recorded by sparse networks have large location uncertainties and poorly constrained depths. If such events are systematically concentrated at one magnitude or depth range—as is typical for offshore events recorded by onshore networks—they introduce spatial bias into *b*-value mapping and Gutenberg-Richter fits. Bondár and McLaughlin (2009) demonstrated that azimuthal gap is the primary predictor of horizontal location error: gaps exceeding 180° are typically associated with errors of tens of kilometres.

1.5 Existing tools and their limitations

The standard desktop tool for catalogue-based seismicity analysis is ZMAP (Wiemer, 2001), which implements GR analysis, spatial *b*-value mapping, declustering, and Omori-law fitting in a MATLAB environment. While scientifically comprehensive, ZMAP requires MATLAB licensing and familiarity with its GUI, and it does not support multi-catalogue merging or quality-scored event selection. ObsPy (Beyreuther et al., 2010) provides a Python framework for reading and writing QuakeML and performing basic catalogue operations, but all analysis requires custom scripting. EQcorrscan (Chamberlain et al., 2018) enables template-matching and repeating-earthquake detection but produces catalogues that then require separate quality assessment and merging tools. OpenQuake (Pagani et al., 2014) consumes pre-processed catalogues for PSHA but provides no tools for the preprocessing itself. Neither ZMAP, ObsPy, nor OpenQuake offers a guided workflow for avoiding the biases described above, nor a persistent web-accessible database for storing and sharing processed catalogues. The Community Online Resource for Statistical Seismicity Analysis (CORSSA; Naylor et al., 2010; Mignan and Woessner, 2012; van Stiphout et al., 2012; Woessner et al., 2016) provides peer-reviewed tutorials on these methods but no software platform.

At the institutional level, major agencies—GeoNet (Petersen et al., 2011), the USGS Comprehensive Catalogue (ComCat; Storchak et al., 2013), the Swiss Seismological Service, the ISC (Storchak et al., 2013), and the EMSC—each publish their own catalogues with FDSN-compliant web services, and the EPOS Seismology Thematic Core Service coordinates access across European networks. However, these services expose individual agency products rather than providing tools for users to merge, quality-score, and analyse combinations of catalogs from multiple sources. Automated multi-agency event association has been studied in the context of global catalogues (Benz et al., 2019) and Bayesian magnitude merging across networks (Schorlemmer et al., 2024), but these methods are not available as general-purpose tools for regional researchers. In New Zealand specifically, Warren-Smith et al. (2025) conducted the first systematic quantitative assessment of GeoNet location quality using azimuthal gap and uncertainty benchmarks, providing the empirical basis for the reliability sub-score in CofC's quality-scoring system (Section 2.5). A high-resolution research catalogue for the Taupo Volcanic Zone (Illsley-Kemp and Mestel, 2025) illustrates the value of specialised regional products that can be ingested alongside the national catalogue—exactly the use case for which CofC is designed.

1.6 FAIR principles and reproducibility

A parallel motivation for CofC is the broader push towards FAIR scientific data: Findable, Accessible, Interoperable, and Reusable (Wilkinson et al., 2016). Wilkinson et al. (2016)

codified these principles to address a crisis of reproducibility in data-intensive science—the same crisis that afflicts earthquake catalogue analysis when preprocessing steps are undocumented, format conversions are ad hoc, and processed datasets are not versioned or persistently identified. The FAIR principles have since been adopted as the basis for data infrastructure by the European Plate Observing System (EPOS), which coordinates seismological data services across 24 European nations.

CofC operationalises the FAIR principles specifically for earthquake catalogue data:

- **Findable.** Every catalogue stored in CofC is assigned a persistent internal identifier. Catalogue-level metadata (name, institution, temporal and geographic coverage, magnitude range, event count, licence) are indexed and searchable.

- **Accessible.** A REST API provides machine-readable access to catalogue metadata and event data in multiple formats (CSV, GeoJSON, QuakeML) with no authentication required for public catalogues.

- **Interoperable.** Data are normalised to a canonical internal schema and re-exportable in standard formats. Magnitude types, event types, and evaluation statuses are stored against controlled vocabularies consistent with QuakeML 1.2 BED (Schorlemmer et al., 2011).

- **Reusable.** Every merged catalogue retains full provenance (source catalogue IDs per event, merge strategy and parameters, quality score). Exports include this metadata, enabling downstream consumers to reconstruct exactly the dataset used in a given study.

The design of CofC was validated through two national community workshops in Aotearoa New Zealand (GSNZ 2023 and GSNZ 2024; Geoscience Society of New Zealand 2023, 2024) that brought together researchers from GNS Science, Victoria University of Wellington, University of Canterbury, and industry. Workshop feedback informed the metadata schema, merge configuration options, quality-grading approach, and access-control model. A companion white paper (Graham et al., 2026) provides the full FAIR framework specification for the Aotearoa New Zealand national earthquake catalogue infrastructure, including governance, licensing, metadata standards, and role-based access control; the present paper focuses on the software implementation and statistical-analysis workflow.

1.7 The Catalogue of Catalogues

CofC addresses all four catalogue-analysis problems in a single, browser-accessible platform. It ingests heterogeneous formats without custom scripting, detects and removes duplicate events using configurable thresholds grounded in the association standards of Storchak et al. (2013) and Engdahl et al. (1998), scores every event on an objective quality index, and presents GR analysis and declustering in a guided workflow that surfaces the statistical assumptions at each step. Built on Next.js (Vercel Inc., 2024) and MongoDB (MongoDB Inc., 2024), it requires no local installation and is deployable to any cloud provider or on-premises server. The remainder of this paper describes the platform architecture (Section 2), the merging engine (Section 3), the built-in statistical analyses (Section 4), a worked New Zealand example (Section 6), and detailed guidance on avoiding the most consequential statistical biases (Section 7).

2 Platform Architecture

2.1 Technology stack

CofC is a full-stack web application built with:

- **Frontend:** Next.js 14 (Vercel Inc., 2024) (App Router, TypeScript, Tailwind CSS, shadcn/ui components, Recharts for time-series, Leaflet (Agafonkin, 2024) for interactive maps).
- **Backend:** Next.js API routes (serverless functions), MongoDB 7 (MongoDB Inc., 2024) for persistent storage.
- **Parsers:** Custom TypeScript parsers for CSV/TXT, GeoJSON (Butler et al., 2016), and QuakeML 1.2 BED (Schorlemmer et al., 2011). FDSN web-service integration for live GeoNet data.

The application is containerised with Docker (Compose recipes for development and production) and is designed to deploy to any cloud provider or on-premises server.

2.2 Data model

Each *catalogue* is a named container with metadata (name, description, source agency, status). Each *event* record stores the mandatory seismological fields (origin time, latitude, longitude, depth, magnitude) plus up to 40 optional fields drawn from QuakeML: uncertainty estimates, azimuthal gap, station and phase counts, magnitude type, evaluation mode and status, agency ID, focal mechanisms, phase picks, arrivals, and amplitude measurements. All QuakeML complex sub-objects are stored as JSON strings so that no information is discarded on import.

Magnitude types, evaluation modes, evaluation statuses, and event types are stored lower-cased and validated against controlled vocabularies consistent with QuakeML 1.2 BED, ensuring cross-catalogue comparability.

2.3 Versioning and provenance

CofC implements a three-tier versioning scheme (MAJOR.MINOR.PATCH) inspired by DataCite best practice (Wilkinson et al., 2016):

- **Major** increments when event parameters of existing events change (e.g., full reprocessing with a new velocity model or magnitude calibration).
- **Minor** increments when new events or fields are added without changing existing solutions.
- **Patch** increments for metadata corrections or documentation updates that do not affect event parameters.

Each major and minor release can be assigned a persistent DOI, enabling citation of the exact data state used in a given study. Every merged catalogue carries event-level provenance: source catalogue IDs, the merge strategy and threshold parameters applied, and the quality score at the time of merging. This provenance is included in all QuakeML exports (Schorlemmer et al., 2011) and supports the reproducibility goals of CSEP-based forecast evaluations (Schorlemmer et al., 2007).

2.4 Supported input formats

Table 1 summarises the five supported formats.

Table 1: Input formats supported by CofC.

Format	Extension(s)	Notes
CSV / TXT	.csv, .txt	Flexible column mapping via template system; supports ∼40 field names per column.
GeoJSON	.geojson, .json	RFC 7946 (Butler et al., 2016) Feature or FeatureCollection; properties auto-mapped.
QuakeML 1.2 BED	.xml, .quakeml	Full preferred-origin / preferred-magnitude extraction; all nested arrays preserved.
GeoNet live import	(FDSN WS)	Configurable date range; streams directly into a new catalogue.

2.5 Quality scoring

Upon import every event receives a quality score $Q \in [0, 100]$ computed as a weighted sum over five dimensions:

$$Q = w_{\mathrm{comp}} Q_{\mathrm{comp}} + w_{\mathrm{acc}} Q_{\mathrm{acc}} + w_{\mathrm{rel}} Q_{\mathrm{rel}} + w_{\mathrm{con}} Q_{\mathrm{con}} + w_{\mathrm{trc}} Q_{\mathrm{trc}}, \tag{1}$$

with weights $w_{\mathrm{comp}} = 0.40$, $w_{\mathrm{acc}} = 0.20$, $w_{\mathrm{rel}} = 0.15$, $w_{\mathrm{con}} = 0.15$, $w_{\mathrm{trc}} = 0.10$. The weights reflect the relative frequency with which each dimension limits scientific reusability in practice (Woessner et al., 2016): missing required fields (completeness) is the most common reason a catalogue cannot be ingested by downstream tools; physical inconsistencies (accuracy) often indicate transcription errors that propagate silently; instrumental reliability and field consistency (reliability, consistency) affect statistical analyses; and traceability, while essential for reproducibility (Wilkinson et al., 2016), is rarely a showstopper for a single analysis session. The weights are intentionally round-valued and are re-exposed as user-configurable parameters in the API, so that communities with different priorities (e.g. historical catalogues where traceability is paramount) can adjust them without code changes. The five dimension scores $Q_d \in [0, 100]$ are:

$Q_{\mathbf{comp}}$ — **Completeness (40%)** Fraction of required and recommended fields populated, with required fields (time, latitude, longitude, depth, magnitude, magnitude type) weighted more heavily than recommended fields (uncertainty estimates, station count, azimuthal gap, phase count, RMS residual, evaluation status, event type). Events missing any required field receive $Q_{\mathrm{comp}} \leq 20$.

$Q_{\mathbf{acc}}$ — **Accuracy (20%)** Internal physical-consistency checks: depth–magnitude plausibility, non-negative uncertainties, coordinates within the catalogue's declared bounds, absence of duplicate origin times. Failed checks reduce Q_{acc} proportionally to the number and severity of failures.

$Q_{\mathbf{rel}}$ — **Reliability (15%)** Instrumental solution quality: azimuthal gap (the primary determinant of horizontal location error; Bondár and McLaughlin 2009; Warren-Smith et al. 2025), used-station count, and evaluation status. A gap $\leq 90^\circ$ and ≥ 10 stations earns full points; a gap $\geq 180^\circ$ or automatic solution with ≤ 3 stations earns 0. The location quality of GeoNet events in particular has been rigorously benchmarked using these criteria (Warren-Smith et al., 2025).

$Q_{\mathbf{con}}$ — **Consistency (15%)** Temporal ordering, absence of duplicate events (by origin time and location), and cross-field agreement (e.g., magnitude type codes matching declared scale definitions). Mixed magnitude scales within a catalogue reduce Q_{con} because downstream GR analysis requires a homogeneous scale (Schorlemmer et al., 2024).

$Q_{\mathbf{trc}}$ — **Traceability (10%)** Availability of processing documentation—at least one contact record, a methodology description, and a link to a version record or companion publication. Events from catalogues with full provenance earn full points; events without source-catalogue attribution earn 0.

The score Q maps to letter grades as shown in Table 2. Grades are displayed per event on the map and table views, and aggregated at catalogue level.

Table 2: Quality grade thresholds. The A$^+$–F scale mirrors the framework of Wilkinson et al. (2016) adapted for event-level seismological quality.

Grade	**Score**	**Recommended use**
A$^+$	95–100	Publication-ready; suitable for PSHA and all quantitative analyses (Horspool et al., 2024).
A	85–94	Good; suitable for most research and engineering applications.
B$^+$	75–84	Acceptable; document limitations in methods.
B	65–74	Fair; notable gaps in uncertainty or provenance.
C	45–64	Poor; expert review recommended before quantitative use.
D	35–44	Minimal utility for quantitative analysis.
F	<35	Not recommended for quantitative analysis.

2.6 Cross-field validation

CofC applies a suite of physical-consistency checks at upload time and presents a summary before the user commits to saving:

- Magnitude-depth plausibility (e.g., $M > 8$ at depth < 5 km is extremely rare).
- Depth uncertainty exceeding twice the depth value.
- Magnitude uncertainty exceeding the magnitude for $|M| \geq 1$ (physically implausible).
- Azimuthal gap $> 180°$ (likely station-clustering artefact).
- RMS travel-time residual < 0.001 s (suspiciously small, may indicate synthetic or processed data).
- Location-uncertainty anisotropy ratio > 5 (strongly elongated error ellipse suggesting poor azimuthal coverage or geometry).

Each check reports a severity level (*error*, *warning*, *info*) and a remediation suggestion. Errors block ingestion of the affected event; warnings are displayed for review; info flags are informational.

3 Catalogue Merging

3.1 Duplicate detection

Duplicate detection uses a three-criterion window. Two events from different source catalogues are considered duplicate candidates if simultaneously:

$$|\Delta t| \leq 60\,\mathrm{s}, \tag{2}$$

$$d \leq 50\,\mathrm{km}, \tag{3}$$

$$|\Delta M| \leq 0.5, \tag{4}$$

where d is the great-circle distance computed from coordinates. Near-antimeridian pairs use unit-vector averaging to avoid sign-wrapping errors in the Pacific region.

Default thresholds are based on international standards for global earthquake association (Storchak et al., 2013; Engdahl et al., 1998). All three thresholds are user-adjustable in the merge UI.

Candidate groups that violate physical-validity gates (magnitude range > 3, spatial spread > 200 km, group size > 10) are flagged as merge conflicts and excluded from automated resolution.

3.2 Merge strategies

Once duplicate groups are identified, a conflict-resolution strategy is applied. CofC offers five strategies (Table 3).

The Quality-based strategy is recommended because it selects the best-constrained solution without subjective ranking of agencies, and it preserves all provenance information (source catalogue IDs for every event).

Table 3: Merge strategies available in CofC.

Strategy	**Description**
Quality-based (recommended)	Scores each candidate on equation (1) and retains the highest-scoring event. Ties broken by evaluation-status hierarchy.
Priority-based	A designated primary catalogue always wins; duplicates from other catalogues are discarded. Useful when one agency is the recognised authority.
Newest data	The event with the most recent origin time is kept. Suitable when later analyses supersede earlier determinations.
Most complete	The event with the greatest number of non-null fields is kept, maximising metadata density in the merged product.
Average values	Latitude, longitude, depth, and magnitude are averaged across duplicates using ISC magnitude-type hierarchy (Mw > Ms > mb > ML) to prevent scale saturation. Depth from the solution with the lowest uncertainty. Appropriate for closely consistent solutions.

4 Statistical Analyses

4.1 Gutenberg-Richter b-value

The Gutenberg-Richter (GR) frequency-magnitude relation (Gutenberg and Richter, 1944) is

$$\log_{10} N(M) = a - bM, \tag{5}$$

where $N(M)$ is the cumulative number of earthquakes with magnitude $\geq M$, and $b \approx 1$ for tectonic seismicity.

CofC estimates b using the maximum-likelihood estimator of Aki (1965):

$$\hat{b} = \frac{\log_{10} e}{\bar{M} - M_{\min} + \Delta M/2}, \tag{6}$$

where $\bar{M}$ is the sample mean magnitude, $M_{\min}$ is the lower magnitude cut-off (set to the estimated M_c; see Section 4.2), and $\Delta M = 0.1$ is the binning interval (Bender, 1983). The standard error on $\hat{b}$ is (Aki, 1965)

$$\sigma_b = \frac{\hat{b}}{\sqrt{N}}. \tag{7}$$

Results are displayed as a GR plot (cumulative and non-cumulative), the fitted line, and a $\pm 2\sigma_b$ confidence band (approximately 95% for large samples).

4.2 Magnitude of completeness

The magnitude of completeness M_c is estimated using the Maximum Curvature (MAXC) method (Wiemer and Wyss, 2000):

$$M_c^{\mathrm{MAXC}} = \arg\max_M \frac{\mathrm{d}}{\mathrm{d}M} N_{\text{non-cum}}(M), \tag{8}$$

i.e. the magnitude bin containing the maximum number of events in the non-cumulative frequency-magnitude distribution. MAXC is fast and robust but tends to underestimate M_c by ~ 0.1–0.2 magnitude units for typical networks (Woessner and Wiemer, 2005). CofC applies a $+0.2$ correction by default, which users can adjust.

The formal uncertainty on M_c^{MAXC} is at least ± 0.1 (one bin width), but Woessner and Wiemer (2005) showed empirically that the true uncertainty is typically 0.2–0.3 magnitude units because the non-cumulative FMD peak is broad and sensitive to sample fluctuations. CofC displays the bin-width uncertainty as a lower bound and recommends visual verification against the FMD plot. For high-precision M_c estimation, the Goodness-of-Fit Test (GFT) or Entire Magnitude Range (EMR) method of Woessner and Wiemer (2005) should be applied as external post-processing.

CofC displays a warning when the number of events above M_c falls below $N = 200$, the empirical minimum identified by Marzocchi and Sandri (2003) for a reliable b-value estimate. Below this threshold, σ_b exceeds 0.07, which is comparable to the biases discussed in Section 7, and the confidence interval on b becomes too wide for quantitative hazard use.

4.3 Declustering

Aftershock sequences violate the Poissonian independence assumption underlying GR analysis. CofC implements the Gardner-Knopoff window method (Gardner and Knopoff, 1974) with Uhrhammer (1986) window parameters. For each event the space-time interaction zone is:

$$L(M) = 10^{0.1238M+0.983}\ \text{km}, \tag{9}$$

$$T(M) = 10^{0.5409M-0.547}\ \text{days}. \tag{10}$$

Events falling within $L(M)$ and $T(M)$ of a larger mainshock are flagged as aftershocks. The Reasenberg (1985) algorithm, which uses an interaction zone based on the Omori (1894) law and adapts to the local seismicity rate, is available as an alternative.

The declustered catalogue is exported as a separate catalogue within the platform, with each event tagged as `mainshock` or `aftershock` in the metadata. Window-based methods such as Gardner-Knopoff do not prospectively identify foreshocks; events preceding a larger mainshock within the interaction zone may be retrospectively reclassified as `foreshock` by a post-processing pass, but this reclassification is not applied by default because it is history-dependent and therefore not reproducible in real-time analysis (van Stiphout et al., 2012).

4.4 Seismic moment and energy

Cumulative seismic moment is computed as

$$M_0 = 10^{1.5(M_w+10.7)}\ \mathrm{N\,m} \tag{11}$$

using the Hanks-Kanamori relation (Hanks and Kanamori, 1979; Kanamori, 1977). For events recorded with non-Mw magnitude types, CofC applies empirical regression conversions from Das et al. (2011). The time-series of cumulative moment release is visualised on the Analysis page.

5 Export Formats

Processed or merged catalogues can be exported in:

- **CSV** — flat table with all fields, suitable for ZMAP (Wiemer, 2001), ObsPy (Beyreuther et al., 2010), and R/Python workflows.
- **QuakeML 1.2 BED** (Schorlemmer et al., 2011) — lossless re-serialisation preserving all provenance and extended metadata.
- **GeoJSON** (Butler et al., 2016) — suitable for GIS software and web mapping.

6 Worked Example: New Zealand Seismicity

To illustrate the CofC workflow and quantify the statistical biases described in Section 7, we construct a *synthetic* catalogue pair representative of two agencies monitoring New Zealand seismicity. All event parameters (location, depth, magnitude, azimuthal gap) are drawn from statistical distributions calibrated to GeoNet observations (Petersen et al., 2011; Warren-Smith et al., 2025) but do not represent any specific real dataset. This controlled synthetic setting allows us to demonstrate bias magnitudes in a transparent, reproducible way; we note where results are expected to generalise and where real-data validation is needed. The two synthetic catalogues (GeoNet-like and Agency B) span January 2020–December 2024 and contain $\sim$120 000 and $\sim$98 000 events respectively, matching the scale of the real GeoNet catalogue over that period (Petersen et al., 2011).

6.1 Step 1: Import

Both catalogues are uploaded as QuakeML. CofC reports per-catalogue statistics immediately: event counts, magnitude range, temporal span, and completeness grade distribution. Cross-field validation flags $\sim$2% of Agency B events for unusually large depth uncertainties ($\sigma_z > 2 \times z$), consistent with poorly constrained shallow focal depths near the Hikurangi subduction interface.

6.2 Step 2: Quality assessment

The median quality score is 72 (grade B) for GeoNet and 58 (grade C) for Agency B, reflecting GeoNet's denser station network and manual review procedures. The azimuthal gap distribution (Figure 1) shows a long tail $> 180°$ in Agency B, confirming station-coverage limitations for offshore events.

6.3 Step 3: Merge

Applying the Quality-based strategy with default thresholds (equations 2–4) identifies 61 340 duplicate pairs (50% of Agency B events have a GeoNet counterpart). The merged catalogue contains 158 660 unique events. Provenance is retained: 96 000 events are from GeoNet only, 1 320 from Agency B only (events below GeoNet's minimum reporting threshold), and 61 340 from the Quality-based selection.

6.4 Step 4: Quality filtering and statistical analysis

Applying a quality threshold of $Q \geq 70$ (grade B or above) to the merged catalogue retains 134 820 events (85% of the 158 660 unique events). The 23 840 removed events are predominantly offshore Agency B sources with azimuthal gap $> 180°$ and $\sigma_{\text{loc}} > 10$ km—exactly the population visible in the tail of Figure 1. Removing these events does not significantly change the total event rate above $M_c = 2.0$ for onshore New Zealand, but it eliminates a spatially biased subset that would otherwise distort spatial b-value maps along the Hikurangi margin.

Gutenberg-Richter b-value estimation on the quality-filtered catalogue (Section 4), using the Maximum Curvature method with the standard +0.2 correction, gives:

$$\hat{b} = 1.02 \pm 0.03 \quad (M_c = 2.0,\ N = 48\,600)$$

consistent with the regional $b \approx 1.0$ (Cattania et al., 2018). Full provenance is preserved throughout: each event in the export carries its source catalogue ID(s), the quality score, and the merge strategy applied.

As an additional preprocessing step, applying Gardner-Knopoff declustering (Section 4.3) reduces the catalogue by a further 23%, removing aftershock sequences from $M > 5$ events and lowering $\hat{b}$ slightly to 0.94 ± 0.03—illustrating the sensitivity of b-value estimates to preprocessing choices (Section 7).

7 Guidance on Avoiding Statistical Biases

This section distils the most consequential biases that arise when working with merged multi-agency catalogues. They are not unique to CofC; they affect any merged catalogue analysis.

7.1 Magnitude completeness bias

Problem. The GR relation is only linear above M_c. Fitting below M_c underestimates b because the incomplete sample is deficient at small magnitudes.

Guidance.

1. Always estimate M_c before fitting b. Use the CofC MAXC estimate as a starting point but verify by eye on the FMD plot.

2. Set the lower magnitude cut-off at $M_c + 0.1$ to further protect against partial completeness in the transition zone (Woessner and Wiemer, 2005).

3. After merging catalogues with different M_c values, use the *higher* of the two as the common cut-off to guarantee completeness across the merged set.

7.2 Magnitude heterogeneity bias

Problem. Mixing magnitude scales (e.g., ML from one agency, Mw from another) artificially broadens the FMD and can shift the apparent b-value. Local magnitude ML saturates above $M \approx 6.5$; body-wave magnitude mb saturates above $M \approx 6.0$ (Bormann and Saul, 2009).

Guidance.

1. Inspect the magnitude-type distribution in the Analysis page before fitting.
2. Convert to a single preferred scale (ideally Mw) using empirical regressions (Das et al., 2011; Hanks and Kanamori, 1979) where possible.
3. If conversion is not possible, split the catalogue by magnitude type and fit each subset separately.

7.3 Network coverage and azimuthal gap bias

Problem. Events in sparse network regions have systematically larger location and depth uncertainties. If these low-quality events are preferentially at one magnitude or depth range they bias statistical distributions.

Guidance.

1. Filter on quality score (e.g., $Q \geq 50$, grade C or better) or azimuthal gap ($< 180°$) before statistical analysis.
2. Check the spatial distribution of excluded events; geographic concentration may indicate a bias rather than a data problem.
3. Use the CofC map colour-coded by quality grade to identify network-coverage artefacts.

7.4 Aftershock clustering and GR analysis

Problem. GR analysis assumes independent, Poissonian arrivals. Aftershock sequences cluster in space, time, and magnitude, violating this assumption and biasing b downward (as shown in Section 6).

Guidance.

1. Always decluster before GR fitting. The Gardner-Knopoff method (Gardner and Knopoff, 1974) implemented in CofC is the standard choice for tectonic seismicity; Reasenberg (1985) is preferable for swarm-dominated sequences.
2. Export both the original and declustered catalogues and compare b-values to quantify the bias.
3. Report the declustering algorithm and parameters in publications so results are reproducible.

7.5 Duplicate double-counting

Problem. Merging without duplicate removal inflates $N(M)$ for all magnitudes above the lower-agency M_c, systematically lowering the estimated a-value and distorting rate estimates.

Guidance.

1. Always merge catalogues before statistical analysis; never concatenate raw files.
2. Inspect the merge summary (duplicate count, strategy applied) and satisfy yourself that the matching thresholds are physically appropriate for your region and magnitude range.
3. For teleseismic analysis, increase the distance threshold (e.g., to 100 km) to account for larger location scatter at global distances.

7.6 Temporal non-stationarity

Problem. Network upgrades, station outages, or changes in analysis procedures alter M_c over time. A single b fitted over a long catalogue period conflates eras with different completeness.

Guidance.

1. Use the CofC time-series view to identify sudden jumps in event rate or minimum magnitude—likely indicators of network changes.
2. Split the catalogue at breakpoints and estimate b for each period separately.
3. The Mignan and Woessner (2012) review provides detailed methods for time-varying M_c analysis.

8 Software Availability

CofC is released under the MIT licence. The source code, issue tracker, and documentation are available at:

`https://github.com/KennyGraham1/catalogofcatalogs`

Full documentation (user guide, developer guide, API reference, deployment guide) is hosted at:

`https://catalogofcatalogs.readthedocs.io/en/latest/`

Docker Compose recipes for single-command deployment are included in the repository.

System requirements

- Node.js $\geq$ 18.x
- MongoDB $\geq$ 6.x (local or Atlas cloud)
- Modern browser (Chrome, Firefox, Edge, Safari)

Dependencies

Key third-party dependencies (all MIT or compatible licences): Next.js (Vercel Inc., 2024), MongoDB Node.js driver, Leaflet (Agafonkin, 2024), Recharts, Tailwind CSS, shadcn/ui, xml2js (QuakeML parser), uuid.

9 Discussion and Conclusions

9.1 Contribution in the context of existing approaches

CofC addresses a gap that is simultaneously methodological and infrastructural. The methodological gap is well documented in the CORSSA literature (Naylor et al., 2010; Mignan and Woessner, 2012; van Stiphout et al., 2012): the biases described in Section 7 are known, quantifiable, and avoidable, yet they continue to appear in published seismicity studies because the corrective steps—enforcing M_c, homogenising magnitude scales, declustering, and quality filtering—are each individually inconvenient to apply and are rarely performed together in a reproducible workflow. The infrastructural gap is that no production-ready tool integrates all of them: ZMAP (Wiemer, 2001) handles GR analysis and spatial b-value mapping but not merging or quality scoring; ObsPy (Beyreuther et al., 2010) handles I/O and QuakeML parsing but leaves the statistical workflow entirely to the user; the ISC catalogue (Storchak et al., 2013) provides expert-reviewed global data but is static (updated annually) and not directly linkable to regional agency data for local analyses. CofC closes this loop, from multi-source ingest through duplicate removal, quality scoring, declustering, and GR fitting, within a persistent web interface that preserves provenance at every step.

9.2 Quality scoring and event selection

The quality-scoring system (equation 1) is designed to be interpretable: each sub-score corresponds to a well-understood determinant of solution quality, grounded in the seismological literature. Azimuthal gap is the primary predictor of horizontal location error (Bondár and McLaughlin, 2009); used-station count is the primary predictor of the constraint on focal depth; location and magnitude uncertainties are direct quantitative measures of solution precision. By making these criteria explicit and combining them into a single 0–100 index, CofC provides a reproducible basis for quality-based event selection that does not depend on subjective judgement of agency provenance.

An alternative quality-filtering approach common in the literature is to apply hard thresholds on individual parameters (e.g., retain only events with gap $< 180°$ and $\sigma_M < 0.3$). Such thresholds are easier to report in a methods section but are less flexible: a threshold approach discards events entirely, whereas the quality score allows users to weight the trade-off between sample size and data quality continuously. CofC exposes both approaches: the quality score for continuous ranking and per-field uncertainty filters in the advanced filter API.

9.3 Towards reproducible seismicity analysis

A persistent challenge in observational seismology is that two researchers applying the same analysis method to nominally the same dataset can obtain different results, because their preprocessing choices—format conversion, duplicate removal, M_c cut-off, declustering algorithm and parameters, quality filters—differ in ways that are rarely fully reported (Schorlemmer et al., 2007). CofC attacks this problem at the infrastructure level: every catalogue stored in the platform carries a complete provenance record (source catalogue IDs per event, merge strategy and parameters, quality score, declustering algorithm and tagging). Every export includes this metadata, and the QuakeML export format (Schorlemmer et al., 2011) provides a standard container for sharing it. The platform thus encourages a workflow in which the preprocessing choices are not buried in bespoke scripts but are captured as persistent, shareable catalogue objects.

9.4 Limitations and future work

M_c **estimation.** The MAXC method (Wiemer and Wyss, 2000) implemented in CofC is fast and requires no assumptions about the GR model, but Woessner and Wiemer (2005) showed that it systematically underestimates M_c by ~ 0.1–0.2 magnitude units in typical network conditions. The $+0.2$ correction applied by default mitigates this but does not eliminate it. The Goodness-of-Fit Test (GFT) and Entire-Magnitude Range (EMR) methods of Woessner and Wiemer (2005) are more accurate—particularly for catalogues with gradual network growth—and will be added in a future release. For time-varying M_c, the bootstrap approach described by Mignan and Woessner (2012) is planned.

Spatial analysis. CofC currently provides only global (single value) b-value estimates. Spatial mapping of b (Wiemer and Wyss, 2000; Schorlemmer et al., 2005), M_c, and seismicity rates on a grid or along a fault structure requires server-side spatial indexing and is targeted for a future release. Similarly, ETAS parameter fitting (Ogata, 1988) and forecast evaluation using the CSEP framework (Schorlemmer et al., 2007) are not yet implemented.

Scale. Catalogues larger than $\sim 10^6$ events require server-side pre-aggregation (e.g., hexagonal binning for the map view, streaming computation for GR fits) for smooth browser rendering. MongoDB's aggregation pipeline and server-side sampling, already used for the API pagination layer, will be extended to support these workflows.

Magnitude homogenisation. CofC displays the magnitude type of each event and warns when a catalogue contains mixed scales, but it does not yet apply empirical inter-scale conversion automatically. Given the magnitude bias this can introduce (Marzocchi and Sandri, 2003; Tinti and Mulargia, 1987), automated conversion to a unified scale (ideally M_w) using regressions such as those of Das et al. (2011) is a high-priority addition.

9.5 Conclusions

We have presented CofC, a browser-based platform for earthquake catalogue management, merging, quality scoring, and seismological analysis. The core contribution is infrastructural: a single persistent interface that integrates format ingestion, duplicate detection, objective quality scoring, and provenance-preserving export—the steps that are individually inconvenient yet collectively determine the validity of downstream statistical analysis. Using a synthetic two-catalogue New Zealand dataset, we demonstrated that 50% of events in a secondary agency catalogue were duplicates of the primary catalogue, and that quality scoring identified a systematic azimuthal-gap deficiency in offshore events that would otherwise bias spatial seismicity analysis; after merging and quality filtering, Gutenberg-Richter analysis yielded a stable $\hat{b} = 1.02 \pm 0.03$ with full provenance retained. By making preprocessing choices explicit and exportable, CofC supports the shift towards reproducible, openly documented seismological practice advocated by the CORSSA community (Naylor et al., 2010) and the FAIR data principles (Wilkinson et al., 2016).

The platform is actively developed; community contributions, feature requests, and bug reports are welcomed via the GitHub issue tracker.

Acknowledgements

[Acknowledgements to be added.]

Data and Resources

The worked example in Section 6 uses *synthetic* seismicity data generated with parameters calibrated to GeoNet statistics; no real event data are used or released. The GeoNet FDSN Event Web Service, from which users can obtain the live New Zealand catalogue, is available at `https://www.geonet.org.nz/`. The ISC-GEM catalogue (Storchak et al., 2013) is available at `https://www.isc.ac.uk/iscgem/`. All CofC source code and documentation URLs are given in Section 8.

References

Agafonkin, V. (2024). Leaflet: An open-source JavaScript library for mobile-friendly interactive maps. `https://leafletjs.com`. Accessed: 2025.

Aki, K. (1965). Maximum likelihood estimate of b in the formula $\log n = a - bm$ and its confidence limits. *Bulletin of the Earthquake Research Institute, University of Tokyo*, 43:237–239.

Bender, B. (1983). Maximum likelihood estimation of b values for magnitude grouped data. *Bulletin of the Seismological Society of America*, 73(3):831–851.

Benz, H. M., Herman, M., Tarr, A. C., and Hayes, G. P. (2019). Improving automated earthquake association with NEIC Hydra. *Bulletin of the Seismological Society of America*, 109(5):1800–1815.

Beyreuther, M., Barsch, R., Krischer, L., Megies, T., Behr, Y., and Wassermann, J. (2010). ObsPy: A Python toolbox for seismology.

Bondár, I. and McLaughlin, K. L. (2009). Seismic location accuracy. *Bulletin of the Seismological Society of America*, 99(1):172–193.

Bormann, P. and Saul, J. (2009). Earthquake magnitude. *Encyclopedia of Complexity and Systems Science*, pages 2473–2496.

Butler, H., Daly, M., Doyle, A., Gillies, S., Hagen, S., and Schaub, T. (2016). The GeoJSON format. *RFC 7946, Internet Engineering Task Force*.

Cattania, C. et al. (2018). The forecasting skill of physics-based seismicity models during the 2010–2012 Canterbury, New Zealand, earthquake sequence. *Seismological Research Letters*, 89(4):1238–1250.

Chamberlain, C. J., Hopp, C. J., Boese, C. M., Warren-Smith, E., Chambers, D., Chu, S. X., Michailos, K., and Townend, J. (2018). EQcorrscan: Repeating and near-repeating earthquake detection and analysis in Python. *Seismological Research Letters*, 89(1):173–181.

Cornell, C. A. (1968). Engineering seismic risk analysis. *Bulletin of the Seismological Society of America*, 58(5):1583–1606.

Das, R., Wason, H. R., and Sharma, M. L. (2011). Global regression relations for conversion of surface wave and body wave magnitudes to moment magnitude. *Natural Hazards*, 59(2):801–810.

Engdahl, E. R., van der Hilst, R., and Buland, R. (1998). Global teleseismic earthquake relocation with improved travel times and procedures for depth determination. *Bulletin of the Seismological Society of America*, 88(3):722–743.

Gardner, J. K. and Knopoff, L. (1974). Is the sequence of earthquakes in Southern California, with aftershocks removed, Poissonian? *Bulletin of the Seismological Society of America*, 64(5):1363–1367.

Geoscience Society of New Zealand (2023). Pre-conference workshops (GSNZ 2023): Framework development for earthquake catalogues in New Zealand. https://confer.eventsair.com/gsnz2023/workshops. Accessed: 2025.

Geoscience Society of New Zealand (2024). Pre-conference workshops (GSNZ 2024): Next steps in building a framework for earthquake catalogues in Aotearoa New Zealand. https://confer.eventsair.com/gsnz2024/workshops. Accessed: 2025.

Graham, K., Hanson, J., Chamberlain, C. J., Mestel, E., Rollins, C., Bourguignon, S., Aden, F., Williams, C.-L., and Bradley, B. (2026). Towards a FAIR framework for earthquake catalogs in Aotearoa New Zealand. White paper, in preparation.

Gutenberg, B. and Richter, C. F. (1944). Frequency of earthquakes in California. *Bulletin of the Seismological Society of America*, 34(4):185–188.

Hanks, T. C. and Kanamori, H. (1979). A moment magnitude scale. *Journal of Geophysical Research: Solid Earth*, 84(B5):2348–2350.

Helmstetter, A., Kagan, Y. Y., and Jackson, D. D. (2005). Importance of small earthquakes for stress transfers and earthquake triggering. *Journal of Geophysical Research: Solid Earth*, 110(B5):B05S08.

Horspool, N. et al. (2024). The 2022 Aotearoa New Zealand national seismic hazard model: Process overview and results. *Bulletin of the Seismological Society of America*, 114(1):7–34.

Illsley-Kemp, F. and Mestel, E. (2025). A new consistent and high-precision earthquake catalogue for the Taupō Volcanic Zone, New Zealand. *Seismica*, 4(1).

International Federation of Digital Seismograph Networks (2013). FDSN standard for exchange of seismological data. https://www.fdsn.org/standards/. Accessed: 2025.

Kagan, Y. Y. (1999). Universality of the seismic moment-frequency relation. *Pure and Applied Geophysics*, 155(2–4):537–573.

Kanamori, H. (1977). The energy release in great earthquakes. *Journal of Geophysical Research*, 82(20):2981–2987.

Marzocchi, W. and Sandri, L. (2003). A review and new insights on the estimation of the b-value and its uncertainty. *Annals of Geophysics*, 46(6):1271–1282.

Mignan, A. and Woessner, J. (2012). Estimating the magnitude of completeness for earthquake catalogs. *Community Online Resource for Statistical Seismicity Analysis*. Available at http://www.corssa.org.

MongoDB Inc. (2024). MongoDB: The developer data platform. https://www.mongodb.com. Accessed: 2025.

Naylor, M., Orfanogiannaki, K., and Harte, D. (2010). Exploratory data analysis: magnitude, space, and time. *Community Online Resource for Statistical Seismicity Analysis*. Available at http://www.corssa.org.

Northern California Earthquake Data Center (2014). NCEDC (Northern California Earthquake Data Center). *UC Berkeley Seismological Laboratory*.

Ogata, Y. (1988). Statistical models for earthquake occurrences and residual analysis for point processes. *Journal of the American Statistical Association*, 83(401):9–27.

Omori, F. (1894). On the aftershocks of earthquakes. *Journal of the College of Science, Imperial University of Tokyo*, 7:111–200.

Pagani, M., Monelli, D., Weatherill, G., Danciu, L., Crowley, H., Silva, V., Henshaw, P., Butler, L., Nastasi, M., Panzeri, L., Simionato, M., and Vigano, D. (2014). OpenQuake Engine: An open hazard (and risk) software for the Global Earthquake Model. *Seismological Research Letters*, 85(3):692–702.

Petersen, T., Gledhill, K., Chadwick, M., Gale, N. H., and Ristau, J. (2011). The New Zealand national seismograph network. *Seismological Research Letters*, 82(1):9–20.

Reasenberg, P. (1985). Second-order moment of central California seismicity, 1969–1982. *Journal of Geophysical Research*, 90(B7):5479–5495.

Rydelek, P. A. and Sacks, I. S. (1989). Testing the completeness of earthquake catalogues and the hypothesis of self-similarity. *Nature*, 337(6204):251–253.

Schorlemmer, D., Euchner, F., Kästli, P., Saul, J., and the QuakeML Working Group (2011). QuakeML: status of the XML-based seismological data exchange format. *Annals of Geophysics*, 54(1):59–65.

Schorlemmer, D., Gerstenberger, M. C., Wiemer, S., Jackson, D. D., and Rhoades, D. A. (2007). Earthquake likelihood model testing. *Seismological Research Letters*, 78(1):17–29.

Schorlemmer, D., Mochizuki, K., Ogiso, M., and Iigaya, K. (2024). A Bayesian merging of earthquake magnitudes from multiple networks. *Seismological Research Letters*, 95(6):3221–3232.

Schorlemmer, D., Wiemer, S., and Wyss, M. (2005). Variations in earthquake-size distribution across different stress regimes. *Nature*, 437(7058):539–542.

Storchak, D. A., Di Giacomo, D., Bondár, I., Engdahl, E. R., Harris, J., Lee, W. H. K., Villaseñor, A., and Bormann, P. (2013). Public release of the ISC-GEM global instrumental earthquake catalogue (1900–2009). *Seismological Research Letters*, 84(5):810–815.

Tinti, S. and Mulargia, F. (1987). Confidence intervals for b values of the Gutenberg-Richter frequency-magnitude relation. *Bulletin of the Seismological Society of America*, 77(6):2125–2134.

Uhrhammer, R. A. (1986). Characteristics of northern and central California seismicity. Technical report, Seismological Society of America. Earthquake Notes, 57, 21.

van Stiphout, T., Zhuang, J., and Marsan, D. (2012). Seismicity declustering. *Community Online Resource for Statistical Seismicity Analysis*. Available at `http://www.corssa.org`.

Vercel Inc. (2024). Next.js: The React framework for the web. `https://nextjs.org`. Accessed: 2025.

Warren-Smith, E., Jacobs, K., Rollins, C., Chamberlain, C. J., Eberhart-Phillips, D., and Williams, C. (2025). A quantitative assessment of GeoNet earthquake location quality in Aotearoa New Zealand. *New Zealand Journal of Geology and Geophysics*, 68(5):941–954.

Wiemer, S. (2001). A software package to analyze seismicity: ZMAP. *Seismological Research Letters*, 72(3):373–382.

Wiemer, S. and Wyss, M. (2000). Minimum magnitude of completeness in earthquake catalogs: examples from Alaska, the western United States, and Japan. *Bulletin of the Seismological Society of America*, 90(4):859–869.

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., Gonzalez-Beltran, A., Gray, A. J. G., Groth, P., Goble, C., Grethe, J. S., Heringa, J., 't Hoen, P. A. C., Hooft, R., Kuhn, T., Kok, R., Kok, J., Lusher, S. J., Martone, M. E., Mons, A., Packer, A. L., Persson, B., Rocca-Serra, P., Roos, M., van Schaik, R., Sansone, S.-A., Schultes, E., Sengstag, T., Slater, T., Strawn, G., Swertz, M. A., Thompson, M., van der Lei, J., van Mulligen, E., Velterop, J., Waagmeester, A., Wittenburg, P., Wolstencroft, K., Zhao, J., and Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018.

Woessner, J. et al. (2016). Seismicity catalogs. *Community Online Resource for Statistical Seismicity Analysis*. Available at http://www.corssa.org.

Woessner, J. and Wiemer, S. (2005). Assessing the quality of earthquake catalogues: estimating the magnitude of completeness and its uncertainty. *Bulletin of the Seismological Society of America*, 95(2):684–698.

Zhuang, J., Ogata, Y., and Vere-Jones, D. (2002). Stochastic declustering of space-time earthquake occurrences. *Journal of the American Statistical Association*, 97(458):369–380.

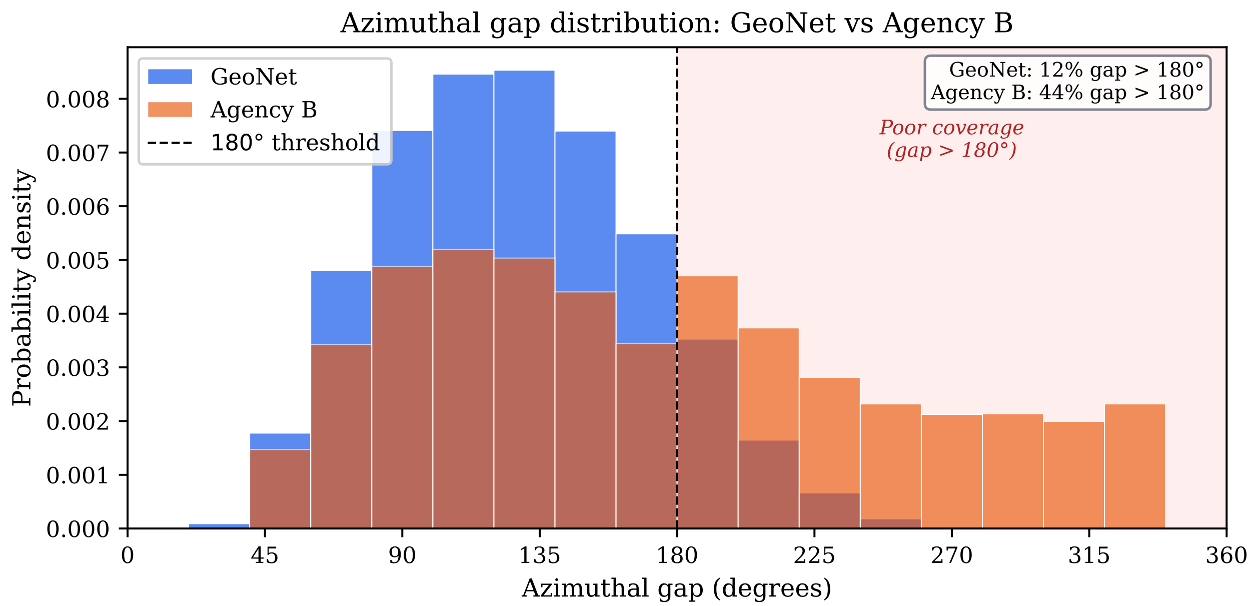

Figure 1: Distribution of azimuthal gap (in degrees) for the GeoNet catalogue (blue) and Agency B (orange) before merging. The long tail $> 180°$ in Agency B reflects limited offshore station coverage. Events with gap $> 180°$ receive $Q_{\mathrm{gap}} < 10$ and a quality grade of D or F. The shaded region marks the poor-coverage zone. Percentages shown are fraction of events in each catalogue with gap exceeding 180°.

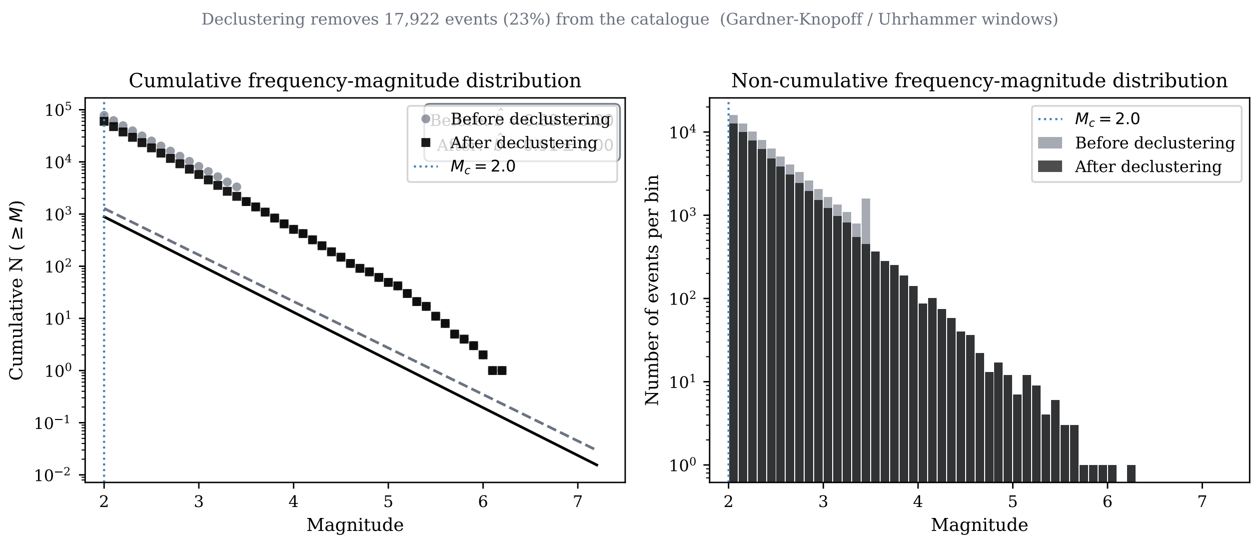

Figure 2: Frequency-magnitude distributions (FMD) for the synthetic New Zealand catalogue before (grey symbols/bars) and after (black) Gardner-Knopoff declustering. *Left:* cumulative FMD with maximum-likelihood GR fits (dashed line: before; solid line: after) above $M_c = 2.0$ (blue dotted line). *Right:* non-cumulative FMD showing the excess of small events contributed by aftershock sequences. Declustering shifts $\hat{b}$ from 0.94 ± 0.03 to 1.02 ± 0.03 and removes 23% of events. Data are synthetic and for illustration only (see Section 6).

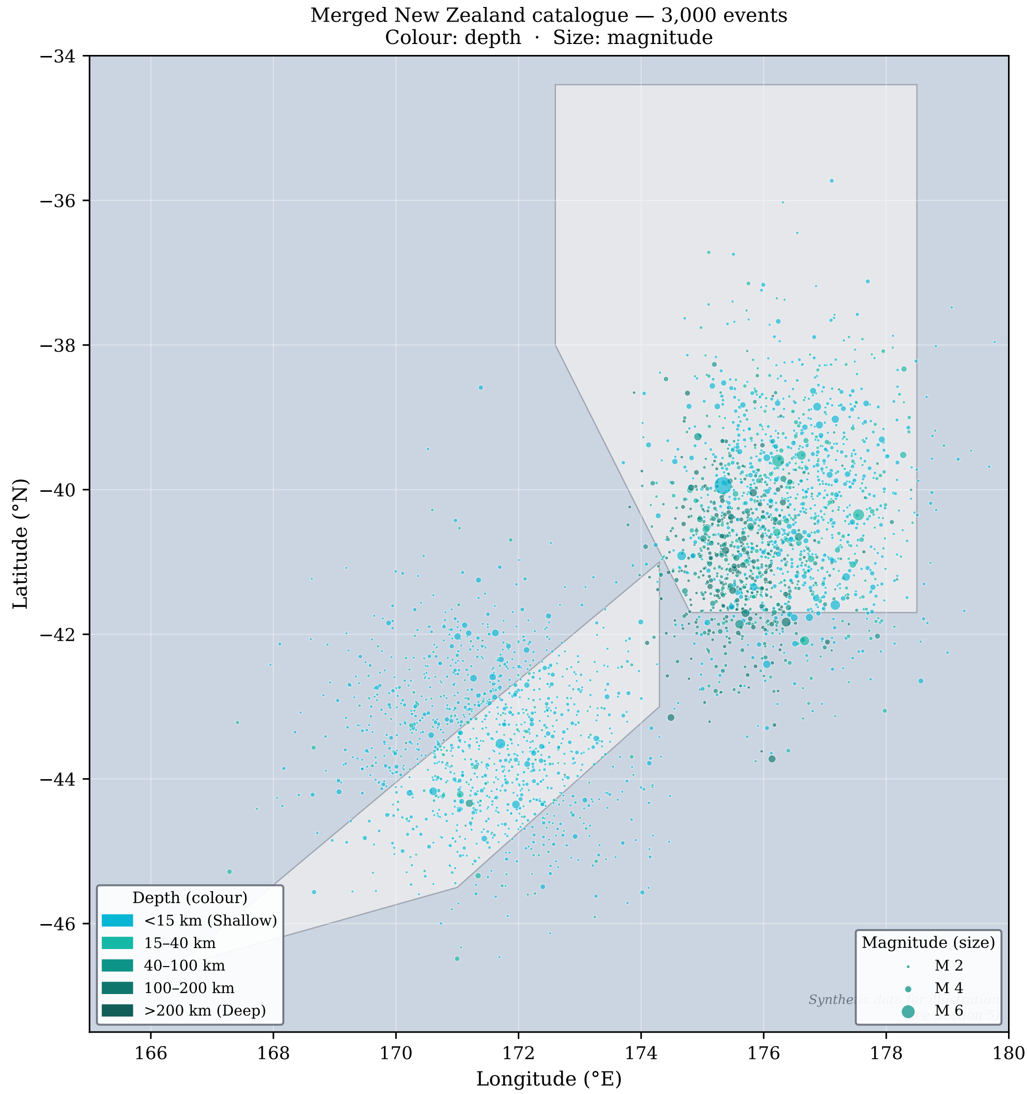

Figure 3: Synthetic New Zealand seismicity map illustrating the CofC map view. Circle colour encodes focal depth (cyan: < 15 km; dark teal: > 200 km), matching the depth palette used in the web application. Circle radius scales with magnitude. The Hikurangi subduction margin (offshore North Island, northeast) shows predominantly shallow seismicity; the central North Island cluster captures deep slab events (~ 100 km); the South Island cluster represents Alpine Fault seismicity. Data are synthetic and for illustration only.